

System Context Diagram — PE Designer Platform


Power Electronics Engineer
A user who designs, simulates, and validates power-electronics circuits (e.g. PCB / magnetics designs).


Administrator
Manages users, reviews uploads, and runs admin tooling (analytics, file editor, simulation uploads).

PE Designer Platform
Lets users build circuit designs, run electrical simulations, get AI-assisted core-loss/magnetics predictions, manage credits, and get help from an AI chatbot.

LTspice (host simulator)
Third-party SPICE simulation engine installed on a host machine; invoked by the simulation worker (Qspice planned).

Ollama
Self-hosted local LLM runtime used as the default chatbot model provider.

Cloud LLM Providers
OpenAI, Anthropic, Google Gemini, OpenRouter — optional alternative chatbot model providers.

Pinecone
Managed vector database used for chatbot RAG (document/knowledge retrieval).

SerpAPI
Third-party web & image search API used by the chatbot's research tools.

Legend
person
system
external person
external system

Designs circuits, runs simulations, views results, chats with AI assistant, manages credits [HTTPS]

Administers users, content, and uploads [HTTPS]

Triggers circuit simulations on [HTTP / host process]

Generates chatbot responses using [HTTP]

Optionally generates chatbot responses using [HTTPS/API]

Stores & retrieves embeddings for RAG using [HTTPS/API]

Performs web/image searches via [HTTPS/API]